Fiscal Year:	FY 2017	**Task Last Updated:**	FY 04/17/2017
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Project Title:	Differential Effects on Telomeres and Telomerase in Twin Astronauts Associated with Spaceflight		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Biomedical countermeasures		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
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Zip Code:	80523-1618	**Congressional District:**	2
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2013-14 HERO NNJ13ZSA002N-TWINS
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No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
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Flight Program:	ISS		
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Task Description:	The ends of human chromosomes are capped by telomeres, tandem arrays of repetitive DNA sequence, which together with associated proteins serve to protect chromosomal termini from inappropriate degradation. Functional telomeres also prevent these natural DNA ends from being recognized as broken DNA (double-strand breaks--DSBs) and triggering inappropriate DNA damage responses, which could result in chromosomal instability. It is well established that telomere length diminishes with cellular division and normal aging. It is also becoming increasingly appreciated that telomere length is influenced by a variety of other factors including life stress, infection, inflammation, and oxidative stress. Nobel Laureate Elizabeth Blackburn suggests that telomere shortening “powerfully quantifies life’s insults.” Indeed, recent research supports telomere length maintenance as a key integrating component for the cumulative effects of genetic, environmental, and lifestyle factors on aging and aging-related diseases. Thus, the rate at which telomeres shorten provides an informative biomarker of biological aging and further, erosion of telomere length can be linked to age-related pathologies including reduced immune function, loss of fertility, cardiovascular disease, and cancer. Healthy lifestyles and diets are positively correlated with telomere length, and changes in diet and lifestyle have also been shown to modulate telomerase activity. Telomerase is the specialized enzyme present in germ line, stem, and cancer cells, capable of de novo addition of telomeric repeats onto the ends of newly synthesized telomeres. A large remaining puzzle in human populations is the relationship between telomerase activity (e.g., measured in peripheral blood lymphocytes) and lifestyle factors (e.g., nutritional, psychological, and physical stressors). We speculate that for the astronauts, telomere length represents a particularly relevant biomarker, because it reflects the combined exposures and experiences encountered during space travel. That is, an individual astronaut’s genetic susceptibilities, exposures to galactic cosmic radiation, physical, and psychological stressors encountered, are all “captured” as changes in telomere length. A remarkable once-in-a-life time opportunity currently exits to evaluate the effects of spaceflight factors on the monozygotic twin astronauts Scott and Mark Kelly, who will have spent a total of 540 and 54 days in low Earth orbit, respectively. We propose to assess differential effects on telomere length and telomerase activity associated with the upcoming yearlong International Space Station (ISS) mission in the twins – individuals matched genetically and who share similar, unique lifestyles; i.e., they are of similar “nature and nurture.” Standard assays will be employed to determine telomere length and telomerase activity as part of their respective and concurrent biodosimetry/chromosome analyses pre-flight (to establish baselines), during flight (to evaluate short-term/temporary changes), and post-flight (to evaluate long-term/permanent changes). Results from the space- and ground-based twins will be compared to evaluate effects of spaceflight factors on these informative biomarkers. Results from the twins will also be compared with our current retrospective analyses of a cohort of unrelated astronauts who have participated in missions of 3 to 6 months duration to evaluate inter-individual sensitivity to these effects of spaceflight, a strategy that will also facilitate evaluation of time dependent responses.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Aging and age-related degenerative pathologies like dementia, cardiovascular disease, and cancer are everyday concerns on Earth as well as in space. Taken together with the other TWINS Study investigations, results from this research will facilitate identification of interactive effects and relationships that will suggest potential mitigation strategies for future study, as well as benefit precision medicine efforts in monitoring general health, and assessing aging and disease risk not only of astronauts, but for those on Earth as well.
Task Progress:	This study is part of an integrated compilation of multiple studies led by multiple Principal Investigators (PIs) at multiple centers that takes advantage of a remarkable and unique opportunity to investigate the effects of space travel with monozygotic twin astronauts, who will have spent a total of ~540 and 54 days in low Earth orbit; importantly, they are of similar “nature and nurture.” For the purposes of the Twins study, the flight subject was on-orbit (ISS--International Space Station) for ~one year, while the ground subject remained Earth-bound for that same time. This particular project will interrogate changes in telomere length and telomerase activity associated with prolonged and combined exposures to galactic space radiations, microgravity, and other spaceflight stressors (e.g., nutritional, psychological, physical). Samples from each of the twins will be assessed for telomere length and telomerase activity at similar times before the flight, during flight, and after return to Earth, to determine whether observed changes differ relative to each other (i.e., are space-flight specific), as well as how they relate to physiological, molecular, genetic, and biochemical changes (information obtained via data sharing with other Twins Study investigations). Telomere length maintenance captures the interplay between genetics and life experiences/exposures, and so is a key integrating component for the cumulative effects of genetic, environmental, and lifestyle factors on aging and age related degenerative diseases, ranging from reduced immune function and dementia, to cardiovascular disease and cancer. Therefore, we propose that for the astronauts, telomere length maintenance represents a particularly relevant biomarker, as it reflects the combined exposures and experiences encountered during space travel. All samples from both Mark and Scott Kelly have now been collected and processed. While some data analyses are continuing, our preliminary results indicate that in contrast to our original hypothesis that accelerated telomere shortening would be associated with the stresses and exposures of spaceflight, telomere length was actually longer in-flight. We are confirming this intriguing result by a variety of means, as well as seeking mechanistic insight via data sharing with the other Twins Study investigators.
Bibliography Type:	Description: (Last Updated: 12/10/2025)
Abstracts for Journals and Proceedings	McKenna MJ, Taylor L, George K, Bailey SM. "Assessing Telomere length and Telomerase Activity in Twin and Unrelated Astronauts." Invited platform presentation at the 2017 Human Research Program Investigators’ Workshop, Galveston TX, January 23-26, 2017. 2017 Human Research Program Investigators’ Workshop, Galveston TX, January 23-26, 2017. Abstract book. , Jan-2017

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